Project Title: Contact Manager

Team ID: SWTID-2026-3620

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Abstract:

Contact Manager is a smart application designed to help individuals and organizations efficiently store, organize, and manage contact information. The system provides a simple interface for adding, editing, and categorizing contacts, along with search and analytics features. With reminders, grouping options, and secure storage, users can maintain better communication and streamline their personal or professional networks.

Introduction:

Managing contacts has become essential in both personal and professional life. Traditional methods like notebooks or basic phone contact lists often lack advanced features such as categorization, reminders, and analytics. Contact Manager aims to provide a digital solution that simplifies contact management while offering intelligent features for better organization and communication.

Problem Statement:

Many people struggle with organizing and retrieving contact information efficiently. Existing solutions may be either too simplistic or overly complex. There is a need for a user-friendly, intelligent system that allows users to manage contacts, categorize them, and receive reminders for important dates or follow-ups.

Objectives:

Provide an easy-to-use interface for storing and managing contacts .Categorize contacts (e.g., family, friends, work, clients).Enable quick search and filtering of contacts .Offer reminders for birthdays, meetings, and follow-ups. Generate insights such as frequently contacted people or communication gaps.

System Overview:

The system processes contact data to generate reports and reminders. It can run as a web or mobile application with cloud storage for secure access.

Key Features

- **Contact Storage** – Add, edit, and delete contacts.
- **Categorization** – Group contacts into categories.
- **Search & Filter** – Quickly find contacts by name, category, or tags.
- **Reminders** – Notifications for birthdays, anniversaries, and meetings.
- **Analytics** – View communication trends and contact frequency.

Technology Stack:

- **Frontend**: React / Flutter
- **Backend**: Python (Flask or FastAPI)
- **Database**: Firebase / PostgreSQL
- **Data Visualization**: Chart.js or similar libraries
- **Cloud Hosting**: AWS / Firebase Hosting

System Modules:

- User Authentication Module
- Contact Management Module
- Categorization & Tagging Module
- Search & Analytics Module
- Reminder & Notification Module

Output:

1. Register API

POST http://localhost:5000/api/auth/register

Body:


```
{
  "name": "Mogana",
  "email": "mogana@test.com",
  "password": "123456"
}
```


Response:


```
{
  "_id": "abc123xyz",
  "name": "Mogana",
  "email": "mogana@test.com",
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI..." }
```

2. Login API

POST http://localhost:5000/api/auth/login

Body:

```
{
  "email": "mogana@test.com",
  "password": "123456"
}
```

Response:

```
{
  "_id": "abc123xyz",
  "name": "Mogana",
  "email": "mogana@test.com",
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI..."
}
```

3. Create Contact

POST http://localhost:5000/api/contacts

Body:

```
{
  "name": "Pavi",
  "phone": "9876543210",
  "email": "pavi@example.com",
  "category": "Work"
}
```

Response:

{
  "_id": "contact123",
  "name": "Pavi",
  "phone": "9876543210",
  "email": "pavi@example.com",
  "category": "Work",
  "createdAt": "2026-03-09T10:00:00Z"
}

4. Get Contacts

GET http://localhost:5000/api/contacts

Headers: Authorization: Bearer <token>

Response:

[
  {
    "_id": "contact123",
    "name": "Pavi",
    "phone": "9876543210",
    "email": "pavi@example.com",
    "category": "Work"
  }
]

5. Update Contact

PUT http://localhost:5000/api/contacts/contact123

Body:

```
{
  "name": "Pavi Updated",
  "phone": "9876543210",
  "email": "pavi.updated@example.com",
  "category": "Work"
}
```

Response:

```
{
  "_id": "contact123",
  "name": "Pavi Updated",
  "phone": "9876543210",
  "email": "pavi.updated@example.com",
  "category": "Work"
}
```

6. Delete Contact

DELETE http://localhost:5000/api/contacts/contact123

Response:

```
{
  "message": "Contact removed"
}
```

Conclusion:

Contact Manager provides a practical and efficient solution for organizing and managing contacts. By combining simple contact storage with intelligent categorization, reminders, and analytics, the system helps users maintain stronger connections and improve communication management.